Machine Learning

29-04-26

* Start linkedin preparation
  - search for jobs like - Data Analyst, Business Analyst
    ML Engineer, AI engineer, etc

  - save those jobs
  - send connection requests to recruiters and employees of those companies

L1 regularization : Lasso regularization

- this regularization can be used in any algorithm which uses gradient descent

eg.

linear regr : $y = w_1x_1 + w_2x_2 + w_3x_3 \ldots w_{10}x_{10} + b$

- when no. of features is large → we'll have to train model for more parameters
  more features = more parameters = complex model = high chances of overfitting

- not all features are important for the task and some are useless but if we include useless features in training → that also causes overfitting

eg. house price prediction

x_1	x_2	x_3	x_4	x_5	y
✓ area	✓ floor	✓ rooms	✓ pincode	✗ apartment name	✓ price

this useless feature for price prediction and if we include this in training, our model will get confused

Hyp funcn: $y = w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + \underline{w_5x_5} + b$

In some cases identifying useless features is easy using domain knowledge

In some cases, even after using domain knowledge, its very difficult to identify useless features.

why we need to deal with useless features?

- as they confuse model
- cause overfiting

$$y = w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + w_5x_5 + w_6x_6 + w_7x_7 + b$$

lets say here we don't know which of these features are useless hence

using regularization we give model ability to find useless features automatically

- After regularization, model will reduce the contribution of useless features in hypothesis function

lets say x_5 and x_7 are not so important /relevant for above hyp. func^n then can try to reduce their contribution in it

↓

this can be done easily if somehow we make $w_5 \approx 0$ and $w_7 \approx 0$

if $w_5 \approx 0$ and $w_7 \approx 0$

$$w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + \underbrace{w_5x_5} + w_6x_6 + \underbrace{w_7x_7}$$

$$\approx w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + w_6x_6$$

this reduces /removes contribution of x_5 and x_7 in model prediction

How to make such weights ≈ 0 automatically?

→ by a small modification in the loss function

Lin regr

$$L = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Lasso regr = Lin regr + lasso regularization

$$L = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \sum_{j=0}^{f} |w_j|$$

n - no. of train data points

f - no. of features in train data with $w_0 = b$

λ - regularization hyperparameter

$$L = \underbrace{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}_{\text{mse}} + \underbrace{\lambda \sum_{j=0}^{f} |w_j|}_{\text{regularization}}$$

mse $\downarrow$ has a minima

lets say we have found minima for <u>mse</u> part

now

we can reduce the value of $\sum |w_j|$ to reduce loss further

$$\sum |w|$$

$$|w_1| + |w_2| + |w_3| + |w_4| + |w_5| + |w_6| + |w_7| + |w_8|$$

$|w_1| \downarrow$ if mse $\uparrow$ $\downarrow$ x_1 is imp

$|w_2| \downarrow$ if mse $\uparrow$ $\downarrow$ x_2 is imp

$|w_5| \downarrow$ if mse change is very minimal $\downarrow$ x_5 is not so important $\downarrow$ $|w_5| \approx 0$

$|w_7| \downarrow$ if mse change is very small $\downarrow$ x_7 is not so imp feature $\downarrow$ $|w_7| \approx 0$

if reducing a weight/model parameter doesn't change mse, it means that parameter and the feature associated with it are not so important.

$$L = MSE + \lambda \sum |w|$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial MSE}{\partial w_1} + \lambda \times 1$$

in lasso regr

$$\frac{\partial L}{\partial w_1} = \frac{\partial MSE}{\partial w_1} + \lambda$$

$\lambda \propto$ strength of regularization

if

λ increases $\rightarrow$ regularization increases $\rightarrow$ overfiting reduces

eg. if $\lambda = 0.01$ is still giving overfiting then we need to increase λ to 0.05 or 0.1 or 0.02

After training with regularization,

$$Y = 2.5x_1 - 1.75x_2 + 0.0002x_3 + 4$$

$w_1 = 2.5$ $\quad w_2 = -1.75$ $\quad w_3 = 0.0002$ $\quad w_0 = b = 4$

order of feature contributions

$$x_1 > x_2 > x_3$$

contribution of feature $\propto$ magnitude of its weight

x_3 is having negligible contribution here as $w_3 \approx 0$

Ridge Regularization :

modified loss function :

$$L = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \sum_{j=0}^{f} (w_j)^2$$

$|w_j| \leftarrow$ L1 regularization = lasso

$w_j^2 \leftarrow$ L2 regularization = ridge

$$\frac{\partial L}{\partial w_1} = \frac{\partial MSE}{\partial w_1} + \lambda \frac{\partial w_1^2}{\partial w_1}$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial MSE}{\partial w_1} + \lambda \times 2 w_1$$

$$w_{1\,new} = w_{1\,old} - \alpha \frac{\partial L}{\partial w_{1\,old}}$$

- derivative of w^2 is smoother than derivative of $|w|$ hence L2 regul. is preferred sometimes over L1

★ Elasticned regularization

$$L = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda_1 \sum_{j=0}^{f} |w_j| + \lambda_2 \sum_{j=0}^{f} (w_j)^2$$

this brings together best of both L1 and L2 regularization